Name: _____

WORKSHEET

Complex number operations

1 Simplify the following expressions.

a $(6 + 2i) + (3 - 5i)$

b $(4 - 5i) - (2 + 2i)$

c $(3 + i) - (4 - 3i)$

d $(5i - 3) + (2 - 3i)$

e $(2 - 3i) - (6 - 2i)$

f $(3 + 6i) - (2i - 3)$

2 Simplify the following expressions.

a $(4 + 3i)(5 - 7i)$

b $(6 + 2i)(3 + 4i)$

c $(5 - i)(5 - 3i)$

d $(4i + 5)(2 + 3i)$

e $(4 - 5i)(6i - 11)$

f $(3 - 2i)(4 + 6i)$

3 Simplify the following expressions.

a $\dfrac{1 - 4i}{3 + 2i}$

b $\dfrac{3 + 2i}{7 + 3i}$

c $\dfrac{2 - i}{4 - 5i}$

d $\dfrac{6 + 3i}{3 - 4i}$

e $\dfrac{i - 5}{2 + 3i}$

f $\dfrac{2 - 3i}{3i - 2}$

4 If $z = -3 + 5i$ and $w = 6 - 2i$, find:

a zw

b $4w - 3z$

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c $(zi - w)(z + wi)$

d $\overline{z}\,\overline{w}$

e $(z + w)^2$

f $\dfrac{zw^2}{\overline{w}}$

5 Simplify each of the following.

a $(2 + 3i)^2 - (3 + 5i)(4 - 2i)$

d $\dfrac{(6 - 4i) + (3 + i)}{(i - 5) - (2 + 2i)}$

b $(6 + 2i)(3 - 5i)(2 + 4i)$

e $\dfrac{i - 2}{3i + 5} - (4 - 6i)^2$

c $\dfrac{4 - i}{(3 + 2i)^2}$

f $\dfrac{5 - 3i}{2i - 4} + \dfrac{3 - 4i}{6i + 2}$

6 Find the value of a and b in each of the following, where a and b are real.

a $a + bi = (i - 4)(2 + 3i)$

d $(7a + 6i) - [(2a - 3bi) - (8 - 5bi)] = 10 + 2i$

b $(a + 2i) - (6 + bi) = 3 + 2i$

e $\dfrac{7 - bi}{2i - a} = 3 + 4i$

c $6a + 1 - (2 - 3b)i = 5i - 3$

f $(2a + bi)(3a - 4bi) = 6 + 2i$

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Answers

1 **a** $9 - 3i$

b $2 - 7i$

c $-1 + 4i$

d $-1 + 2i$

e $-4 - i$

f $6 + 4i$

2 **a** $41 - 13i$

b $10 + 30i$

c $22 - 20i$

d $-2 + 23i$

e $-14 + 79i$

f $24 + 10i$

3 **a** $\dfrac{-5 - 14i}{13}$

b $\dfrac{27 + 5i}{58}$

c $\dfrac{13 + 6i}{41}$

d $\dfrac{6 + 33i}{25}$

e $\dfrac{-7 + 17i}{13}$

f -1

4 **a** $-8 + 36i$

b $33 - 23i$

c $22 - 120i$

d $-8 - 36i$

e $18i$

f $\dfrac{76 + 168i}{5}$

5 **a** $-27 - 2i$

b $152 + 64i$

c $\dfrac{8 - 53i}{169}$

d $\dfrac{-6 + 3i}{5}$

e $\dfrac{673 + 1643i}{34}$

f $\dfrac{-35 - 11i}{20}$

6 **a** $a = -11, b = -10$

b $a = 9, b = 0$

c $a = -\frac{2}{3}, b = \frac{7}{3}$

d $a = \frac{2}{5}, b = 2$

e $a = -5, b = -26$

f $a = \pm 1, b = \dfrac{\pm 5 \pm \sqrt{57}}{8}$

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